

3. Yash in Rwanda (20 marks)

Problem designed by Yash Sinha

Yash, who is studying linguistics in the University of Chicago, went to Rwanda on a field trip. His task was to analyze the Kinyarwanda language, which is one of the three official languages of Rwanda along with French and English. What flummoxed him though is while some English sentences had only one translation in Kinyarwanda, some had two and some even more. To crack the mystery, Yash collected the Kinyarwanda translations of a few English sentences from the locals, though he could not make his subjects speak out all the possible translations (after all they were not linguists and said whatever came to their mind). Yash prepared the following table of Kinyarwanda sentences and their English sentences and also made some notes of how many translations he thought the English sentence actually had for cases where he did not get all possible translations.

1.	<i>Umugabo arakora</i>	The man is working.
2.	<i>Abogore baragenda</i>	The women are going.
3.	<i>Umugabo arabona umugore</i>	The man is seeing the woman.
4.	<i>Umugore arabonwa numugabo</i>	
5.	<i>Umuhungu arabonwa nabogore</i>	The women are seeing the boy.
6.	<i>Abogore barabona umuhungu</i>	
7.	<i>Abogabo baraandika ibaruwa nikarumu</i>	The men are writing the letter with the pen. [There are 2 more possible translations in Kinyarwanda]
8.	<i>Abogabo baraandikiisha ikarumu ibaruwa</i>	
9.	<i>Umuhungu arabona abogore nijisho</i>	The boy is seeing the women with the eye. [there are 2 more possible translations in Kinyarwanda]
10.	<i>Ijisho araboniishwa abogore numuhungu</i>	
11.	<i>Umualimu araandikera umuhungu igitabo</i>	The teacher is writing the book for the boy. [there are 2 more possible translations in Kinyarwanda]
12.	<i>Umualimu araandika igitabo kuumuhungu.</i>	

Yash realized that for every pair of Kinyarwanda sentences that mean the same thing, the two forms mean the same thing. *However, they do not necessarily have the same structure.*

Assignment 1: Explain how the same meaning can be expressed in multiple ways.

Assignment 2 [Answer-booklet]: Translate the following sentences into English

- a) *Ibaruwa araandikwa nabogabo*
- b) *Ibaruwa araandikwa numugore kuabogabo*
- c) *Abogabo barabonera umugore abohungu*

Assignment 3 [Answer-booklet]: Now translate the two sentences below. Consider the verb form carefully.

- d) *Ikarumu araandiishwa ibaruwa numagabo*
- e) *Abogore barabonerwa abohungu numugabo*

Assignment 4 [Answer-booklet]: Provide Kinyarwanda translations for these two English sentences. See instructions besides each sentence for the number of translations you are expected to provide.

- f) The boys are writing the book. (Give **two** Kinyarwanda equivalents)
- g) The boy is writing the book for the women. (Give **four** Kinyarwanda equivalents)